Luiz A. Silva-Lima

Luiz Albérico da Silva Lima Ph.D. in Astrophysics

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Research Experience

Postdoctoral researcher — Minas Gerais, Brazil
National Laboratory for Astrophysics (LNA) — MCTI — 2026–present

Relevant Research Skills

Integral-field spectroscopy analysis. Development of an IFS analysis pipeline based on pPXF, including stellar continuum fitting, stellar kinematics, stellar population properties, and emission-line analysis.

Stellar populations and star-formation histories. Recovery of star-formation histories through regularised fitting of simple stellar population models, including analysis of SFH products from large MUSE datasets.

Multi-component emission-line fitting. Spatially resolved emission-line analysis using multi-component Gaussian modelling (up to three components), with strategies to mitigate local minima and overfitting through statistical model selection.

Galaxy kinematics and dynamics. Analysis of stellar and ionised-gas kinematics using harmonic decomposition with KINEMETRY, bisymmetric flow modelling with DISKFIT, and photometric modelling with Multi-Gaussian Expansion (MGE).

Ionised gas properties. Derivation of spatially resolved gas-phase metallicities using empirical and theoretical calibrations, and electron densities from emission-line ratios.

Stellar spectroscopy. Determination of stellar atmospheric parameters, including effective temperature, surface gravity, and chemical abundances, using FERRE.

Development of astronomical analysis tools. Development and maintenance of SPECTCUBE, development of an SFH and emission-line fitting pipeline, and contributions to nGIST in the GECKOS survey.

Education

PhD in Astrophysics and Computational Physics — São Paulo, Brazil
Universidade Cidade de São Paulo — 2021–2025
Study of the AGN feeding process through bars with the MUSE
Supervisor: Dr. Lucimara P. Martins
Co-supervisor: Dr. Dimitri A. Gadotti

Research visit — Durham, United Kingdom
Durham University — 2023
Supervisor: Dr. Dimitri A. Gadotti

MSc in Astrophysics and Computational Physics — São Paulo, Brazil
Universidade Cruzeiro do Sul — 2019–2021
The Role of Bars in AGN Feeding
Supervisor: Dr. Lucimara P. Martins

BSc in Electrical Engineering — São Paulo, Brazil
Universidade Cruzeiro do Sul — 2014–2018
- With Honours (1st Class).
- **Undergraduate research project** (2016–2018), São Paulo, Brazil. Title: *The Role of Bars in AGN Feeding*.
- **Undergraduate internship** (2018), **Nuclear and Energy Research Institute (IPEN)**, São Paulo, Brazil.

Complementary Training

6th Indo-French Astronomy School: Treasures in the Voxels (remote) — Lyon, France
Centre de Recherche Astrophysique de Lyon — 2020

XXIV Special Course Cycles — Rio de Janeiro, Brazil
National Observatory — 2019

First Light: Stars, galaxies and black holes in the epoch of reionization — São Paulo, Brazil
Universidade de São Paulo — 2019

Grants and Fellowships

PhD fellowship 2021–2025
Coordination of Superior Level Staff Improvement (CAPES)

Research visit at Durham University, six months 2023
National Council for Scientific and Technological Development (CNPq)

MSc fellowship 2019–2021
São Paulo Research Foundation (FAPESP)

Undergraduate research fellowship 2016–2018
National Council for Scientific and Technological Development (CNPq)

Research Interests

My research focuses on galaxy evolution through the interplay between stellar populations, gas dynamics and galaxy structure. I am particularly interested in the role of bars in driving gas inflows, central star formation and AGN fuelling, as well as the processes governing the assembly and evolution of galaxies across cosmic time.

My research combines spatially resolved spectroscopy and stellar population analysis to investigate the properties and kinematics of stars and ionised gas. I have extensive experience with integral-field spectroscopy, particularly MUSE, as well as SDSS data, developing and applying analysis tools for large spectroscopic datasets.

Research Collaborations

GECKOS *Full Member*

Generalising Edge-on galaxies and their Chemical bimodalities, Kinematics, and Outflows out to Solar environments, PI: Jesse van de Sande. 2023–present

- I provide regular nGIST updates to the star formation history recovery procedures, including general improvements to the pipeline.
- I also test the fitting configurations for major data releases and assess the treatment of emission lines during the fits.

TIMER *Associate Member*

Time Inference with MUSE in Extragalactic Rings, PI: Dimitri Gadotti. 2023–present

- I am working with the emission line measurements in the IFS data, and I am leading the effort in the use of multiple Gaussian components.

Technical Skills

Programming languages: Python (advanced), R, C, IDL (basic)
GitHub: GitHub/luizsl

- SPECTCUBE (developer) PyPI | GitHub
- SFH and emission-line fitting pipeline (developer) GitHub
- nGIST (contributor) ADS | ASCL | GitHub

Publication Metrics

Source: NASA ADS public library.

- Records in ADS library = 13 (8 marked refereed).
- Total citations = 135; h-index = 6.
- ADS reads = 3 520.

Referees

Prof. Lucimara P. Martins, Universidade Cidade de São Paulo
lucimara.martins@cruzeirodosul.edu.br

Prof. Dimitri A. Gadotti, Durham University
dimitri.a.gadotti@durham.ac.uk

Prof. Marie Martig Liverpool John Moores University
m.martig@ljmu.ac.uk

Refereed Publications

1. Geraldo Gonçalves, Paula R. T. Coelho, **Luiz A. Silva-Lima**, Lucimara P. Martins, Marie Martig, Dimitri A. Gadotti, Camila de Sá-Freitas, Jairo Mendez-Abreu, Patricia Sánchez-Blázquez, Jesús Falcón-Barroso, Glenn van de Venn, Adriana de Lorenzo-Cáceres, and Isabel Pérez. *Model-dependent metallicities of young stellar populations in TIMER galaxies*. Monthly Notices of the Royal Astronomical Society, Advance Access, October 2026. DOI: 10.1093/mnras/stag1939 (ADS).

2. Elliott, R., Fisher, D. B., Mazzilli Ciraulo, B., Fraser-McKelvie, A., Hayden, M. R., Martig, M., van de Sande, J., Battisti, A. J., Bland-Hawthorn, J., Bolatto, A. D., Brown, T. H., Catinella, B., Combes, F., Cortese, L., Davis, T. A., Emsellem, E., Gadotti, D. A., Pinna, F., Puzia, T. H., **Silva-Lima, L. A.**, Valenzuela, L. M., and van de Ven, G. *The GECKOS Survey: extraplanar ionized gas in star-forming galaxies from eDIG to galaxy-scale winds*. Monthly Notices of the Royal Astronomical Society, 548(3), stag672, 2026. DOI: 10.1093/mnras/stag672 (ADS).

3. B. Mazzilli Ciraulo, D. B. Fisher, R. Elliott, A. Fraser-McKelvie, M. R. Hayden, M. Martig, J. van de Sande, A. J. Battisti, J. Bland-Hawthorn, A. D. Bolatto, T. H. Brown, B. Catinella, F. Combes, L. Cortese, T. A. Davis, E. Emsellem, D. A. Gadotti, C. del P. Lagos, X. Lin, A. Marasco, E. Peng, F. Pinna, T. H. Puzia, **L. A. Silva-Lima**, L. M. Valenzuela, G. van de Ven, and J. Wang. *The GECKOS Survey: resolved, multiphase observations of mass loading and gas density in the galactic wind of NGC 4666*. Monthly Notices of the Royal Astronomical Society, 544(4), 3290–3311, 2025. DOI: 10.1093/mnras/staf1875 (ADS).

4. A. Fraser-McKelvie, J. van de Sande, D. A. Gadotti, E. Emsellem, T. Brown, D. B. Fisher, M. Martig, M. Bureau, O. Gerhard, A. J. Battisti, J. Bland-Hawthorn, A. Boecker, B. Catinella, F. Combes, L. Cortese, S. M. Croom, T. A. Davis, J. Falcón-Barroso, F. Fragkoudi, K. C. Freeman, M. R. Hayden, R. McDermid, B. Mazzilli Ciraulo, J. T. Mendel, F. Pinna, A. Poci, T. H. Rutherford, C. de Sá-Freitas, L. A. Silva-Lima, L. M. Valenzuela, G. van de Ven, Z. Wang, and A. B. Watts. *The GECKOS survey: Identifying kinematic sub-structures in edge-on galaxies*. Astronomy & Astrophysics, 700, A237, 34 pp., 2025. DOI: 10.1051/0004-6361/202452891 (ADS).

5. **Luiz A. Silva-Lima**, Dimitri A. Gadotti, Lucimara P. Martins, Tutku Kolcu, Paula R. T. Coelho, Francesca Fragkoudi, Taehyun Kim, Camila de Sá-Freitas, Jesús Falcón-Barroso, Adriana de Lorenzo-Cáceres, Jairo Méndez-Abreu, Justus Neumann, Miguel Querejeta, and Patricia Sánchez-Blázquez. *Optimizing the analysis of emission lines in galaxies: the case of the MUSE TIMER galaxy NGC 613*. Monthly Notices of the Royal Astronomical Society, Volume 540, Issue 3, pp. 2787–2809, 2025. DOI: 10.1093/mnras/staf837 (ADS).

6. Taehyun Kim, Dimitri A. Gadotti, Miguel Querejeta, Isabel Pérez, Almudena Zurita, Justus Neumann, Glenn van de Ven, Jairo Méndez-Abreu, Adriana de Lorenzo-Cáceres, Patricia Sánchez-Blázquez, Francesca Fragkoudi, Lucimara P. Martins, **Luiz A. Silva-Lima**, Woong-Tae Kim, and Myeong-Gu Park. *Impacts of Bar-driven Shear and Shocks on Star Formation*. The Astrophysical Journal, 968(2), 87, 2024. DOI: 10.3847/1538-4357/ad410e (ADS).

7. Baratella, M., Prabhu, D. S., **Lima, L.**, and Prugniel, P. *Prospects of measuring a metallicity trend and spread in globular clusters from low-resolution spectroscopy*. Astronomy & Astrophysics, 661, A138, 2022. DOI: 10.1051/0004-6361/202142667 (ADS).

8. **Silva-Lima, L. A.**, Martins, L. P., Coelho, P. R. T., and Gadotti, D. A. *Revisiting the role of bars in AGN fuelling with propensity score sample matching*. Astronomy & Astrophysics, 661, A105, 2022. DOI: 10.1051/0004-6361/202142432 (ADS).

Publications Submitted or in Preparation

1. J. Hernández-Yévenes et al. *The GECKOS survey: Resolving the molecular and ionised gas in the galactic outflow of ESO 484-036*. Submitted to MNRAS; arXiv:2604.14546; DOI: 10.48550/arXiv.2604.14546 (ADS).

2. Kolesnikov, I. et al. *Beyond the Disc–Spheroid Dichotomy at $0.6 < z < 8.0$: Evidence for a Morphological Indeterminacy Regime from JWST*. Submitted to MNRAS; arXiv:2610.24078; DOI: 10.48550/arXiv.2610.24078 (ADS).

3. Silva-Lima, L. A. and the TIMER collaboration. *Ionised gas kinematics in the centres of barred and unbarred galaxies with MUSE*. In preparation for submission in 2026.

Software Register

Fraser-McKelvie, A. et al. *nGIST: The new Galaxy Integral-field Spectroscopy Tool*. Astrophysics Source Code Library, ASCL:2507.015, 2025; no DOI assigned (ADS).

Events, Conferences and Workshops

Talks, seminars and posters

2026	Talk	Annual Meeting of the Brazilian Astronomical Society: Kinematics and Excitation in Barred and Unbarred Galaxies	Rio Grande do Norte, BR
2025	Talk	IX NAT Workshop: Emission-Line Properties of NGC 613: Evidence for Multiple Ionisation Mechanisms with MUSE	São Paulo, BR
2024	Talk	Annual Meeting of the Brazilian Astronomical Society: Stellar Population and Emission-Line Properties of NGC 613	São Paulo, BR
2024	Talk	XVIII NAT Workshop: Exploring Ionisation Mechanisms and Stellar Populations in NGC 613	São Paulo, BR
2023	Talk	TIMER Meeting: AGN in TIMER	Granada, ESP
2023	Talk	Special Seminar, Centre for Extragalactic Astronomy (Durham University): Revisiting the Role of Bars in AGN Fuelling	Durham, UK
2023	Talk	III NAT Lectures on Astrophysics: Full Spectral Fitting for the Derivation of Stellar Population and Ionised Gas Properties	São Paulo, BR
2020	Talk	Re-assessment of the Metallicity Trend on the RGB of NGC 6397 (Remote)	Lyon, FR
2019	Talk	II NAT Workshop: The Role of Bars in AGN Feeding	São Paulo, BR
2019	Talk	I NAT Workshop: The Role of Bars in AGN Feeding	São Paulo, BR
2018	Talk	XXII Scientific Initiation Meeting (ENIC): The Role of Bars in AGN Feeding	São Paulo, BR
2017	Talk	XLI Annual Meeting of the Brazilian Astronomical Society: The Role of Bars in AGN Feeding	São Paulo, BR
2017	Talk	XXI Scientific Initiation Meeting (ENIC): The Role of Bars in AGN Feeding	São Paulo, BR
2016	Talk	XX Scientific Initiation Meeting (ENIC): The Role of Bars in AGN Feeding	São Paulo, BR
2015	Talk	XIX Scientific Initiation Meeting (ENIC): The Role of Bars in AGN Feeding	São Paulo, BR
2023	Poster	Galactic Bars: Driving and Decoding Galaxy Evolution: The Role of Bars in AGN Feeding	Granada, ESP
2022	Poster	XLV Annual Meeting of the Brazilian Astronomical Society: Nuclear Structures in the Barred Galaxy NGC 613	Virtual
2021	Poster	XLIV Annual Meeting of the Brazilian Astronomical Society: Bars Feeding AGN: The Need of a Secondary Mechanism	Virtual
2019	Poster	XLIII Annual Meeting of the Brazilian Astronomical Society: An Unbiased Sample of Galaxies for the Study of Bar-Nuclear Activity	São Paulo, BR
2018	Poster	XLII Annual Meeting of the Brazilian Astronomical Society: The Role of Bars in AGN Feeding	São Paulo, BR

Workshops and hack weeks

2026	Hack Week	HUB2026: Hyperspectral Unmixing in Bahia	Bahia, BR
2022	Workshop	Large-Volume Spectroscopic Analyses of AGN and Star-Forming Galaxies in the Era of JWST	Baltimore, US (Remote)
2021	Workshop	AIA: Artificial Intelligence in Astronomy	Virtual
2021	Workshop	Precision Spectroscopy: Stellar Connections, from Galaxy Evolution to Exoplanets	Virtual
2020	Workshop	First Light Virtual Reunion	Virtual

Other events

2026	Round-table	Pesquisa, Cooperação Acadêmica e Impacto: Experiências que contribuem para os ODS	Virtual